

Pwned: The Risk of Exposure From Data Breaches*

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Abstract

News about massive data breaches is increasingly common. But what proportion of Americans are exposed in these breaches is still unknown. We combine data from a large, representative sample of American adults ($n = 5,000$), recruited by YouGov, with data from *Have I Been Pwned* to estimate the lower bound of the number of times Americans' private information has been exposed. We find recorded exposure for 82.84% of sampled adults, averaging 2.99 records per person for the email queried. The better educated, the middle-aged, and women have higher average recorded counts; Whites and Blacks have higher counts than Hispanics/Latinos.

CCS Concepts

• Security and privacy → Social aspects of security and privacy.

Keywords

Security risk, Privacy risk, Data breaches, Digital divide

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1 Introduction

On the Internet, nobody knows you're a dog. So the adage goes. But increasingly, others know that you like dog food and hate cats. Many of us have made our peace with this new reality. A slew of massive account breaches in recent years, however, threatens to pull the rug from under all illusions of anonymity [9].¹

*Research revision, September 10, 2026, of the 2019 paper. Data, scripts, and supporting information can be downloaded from <http://github.com/themains/pwned>.

¹On September 22, 2016, for instance, Yahoo! revealed that 500M accounts had been compromised in a breach [4]. Less than three months later, on December 14, 2016, Yahoo! announced that data had been stolen from nearly 1B user accounts in a different breach [11]. In all, Wikipedia lists 272 separate breaches between 2004 and 2018 (see https://en.wikipedia.org/wiki/List_of_data_breaches)

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But there is little existing research on how frequently Americans' private information is part of such breaches. Much of the research on data breaches has focused on the downstream impact on corporations, e.g., [5, 14, 16], and people, e.g., [2, 3, 10]. Such research is vital—it informs data breach notification policies, e.g., [6, 8, 12]. But absent from the literature is data that is important for developing effective public policy on corporate liability for data breaches—data on the average American's risk of their private information being exposed in a data breach. In this note, we shed light on this question.

Using a unique dataset, we estimate the lower bound of the average number of recorded exposures per person. We merge data from a large representative sample from YouGov ($n = 5,000$) with data from *Have I Been Pwned* (HIBP). We check whether the email associated with the YouGov account is part of the 293 public breaches cataloged by HIBP. We also study how exposure to breaches varies by socio-economic factors including ethnicity, sex, age, and education.

2 Data and Methods

In July 2018, YouGov drew a nationally representative sample of 5,000 adult Americans. YouGov draws the sample as follows: it starts with a random sample of a high-quality sample of American adults, e.g., Current Population Survey, and then finds people on its panel that match the drawn sample most closely [13]. Some research suggests that the quality of samples drawn by YouGov is comparable to those drawn using probability sampling [1]. The lookup required no new survey response: YouGov used the emails associated with the accounts to collect the data. Prior panel participation and coverage still affect generalization to American adults. (YouGov never shared the emails with us.) Table 1 presents the marginals on key socio-demographic variables (see here for the codebook). (Table Supporting Information (SI)² 1.1 presents the comparison between the Current Population Survey (CPS) and YouGov on key demographic variables. The upshot is that on key marginals, differences in age, sex, and education shares are less than five percentage points. Race and ethnicity categories are not commensurate, as discussed in the SI.)

After drawing the sample, YouGov used the emails associated with the accounts to query the HIBP API. (YouGov did the lookups so that it didn't have to share the email IDs.) HIBP is a non-profit clearinghouse of information about online account breaches. HIBP's stated aim is to provide a way for people to check if they are at risk from online breaches. The archived July 2018 catalogue contains data from 293 breaches covering 278 unique domains and

²Supporting information can be downloaded from <http://github.com/themains/pwned>.

Table 1: YouGov Sample Characteristics

	Proportion
Age	
18 to 25	0.15
26 to 35	0.18
36 to 50	0.23
51 to 65	0.26
66 to 100	0.18
Education	
No HS	0.07
HS Grad.	0.33
Some College	0.20
2-year College Degree	0.11
4-year College Degree	0.18
Postgrad Degree	0.11
Sex	
Female	0.51
Male	0.49
Race	
White	0.64
Black	0.12
Hispanic/Latino	0.15
Asian	0.03
Native American	0.01
Mixed Race	0.03
Other	0.02
Middle Eastern	0.00

5,235,843,322 accounts, including data from prominent breaches like the two Yahoo! breaches covering nearly 1.5 billion accounts. The HIBP data are, however, not comprehensive. Security researchers believe that there are many breaches that the companies are unaware of and at least a few cases where a company doesn't share information about a breach it knows about. HIBP also refuses to provide data on sensitive breaches—breached accounts where a person's inclusion may adversely affect them—from their public API³. So data from HIBP only gives us a lower bound.

HIBP provides an easy way to get all the breached accounts associated with a particular email ID—you just need to make a simple API call passing the email that you want to get data on. This method gives us data on all the breaches logged by HIBP for all the 5,000 profiles. There is one caveat. Our YouGov sample provides data associated with only one email ID, the email people used to register with YouGov. People often have multiple email IDs. And that is another reason why all we get from this data is a lower bound. The actual number of exposures per person is likely much higher.

³HIBP website notes that it does not share whether or not an account has been part of the breach at Adult Friend Finder, Ashley Madison, Beautiful People, Bestialitysextaboo, Brazzers, CrimeAgency vBulletin Hacks, Fling, Florida Virtual School, Freedom Hosting II, Fridae, Fur Affinity, HongFire, Mate1.com, Muslim Match, Naughty America, Non Nude Girls, Rosebutt Board, The Candid Board, The Fappening, xHamster and 1 more.

With each request, HIBP returns some metadata on the kind of breaches. (See the codebook for details about all the data that it returns.) Two pieces of information are material here. HIBP classifies each breach as verified or unverified. And it defines unverified breaches as breaches whose “legitimacy” it cannot “establish beyond [a] reasonable doubt.” HIBP includes these unverified breaches because “they still contain personal information about individuals who want to understand their exposure on the web.” The other material column that HIBP returns relates to whether a breach is part of a “spam list.” HIBP defines *SpamList* as cases where “large volumes of personal data are found being utilized for the purposes of sending targeted spam.” HIBP adds, “This often includes many of the same attributes frequently found in data breaches such as names, addresses, phones numbers and dates of birth. The lists are often aggregated from multiple sources, frequently by eliciting personal information from people with the promise of a monetary reward.” And the reason HIBP includes these data is: “whilst the data may not have been sourced from a breached system, the personal nature of the information and the fact that it's redistributed in this fashion unbeknownst to the owners warrants inclusion here.”

3 Results

After removing 28 exact duplicate records, 14,951 recorded exposures are associated with the 5,000 emails on file. On average, there are 2.99 records per person; the median is 3. At least one exposure is recorded for 82.84% of sampled adults. Each record represents one person–breach pair, not a distinct breach incident.

The association between exposure counts and socioeconomic characteristics runs counter to what focusing on traditional concerns around the digital divide would lead us to believe. One possible explanation is that people who use online services more accumulate more exposures. We do not measure service use here, so this remains a hypothesis. (See SI 1 Tables and SI 2 Figures for corresponding regressions and figures illustrating group-wise means along with the 95% confidence intervals.)

The number of recorded exposures increases roughly monotonically with education (see Table SI 1.4 and Figure SI 2.3). The average number of recorded exposures among people with no high school degree is 2.35. Compare this to postgraduates, who are part of 3.20 recorded exposures on average.

In contrast to the relationship between education and the number of recorded exposures, the relationship between the number of recorded exposures and age is curvilinear (see Table SI 1.5), with young people and seniors having lower recorded counts than middle-aged adults. But, as the loess illustrates (see Figure SI 2.4), the relationship is modest.

When we compare the average number of recorded exposures among men and women, we find that women have 1.12 times the average recorded count of men (see Table 2 and Table SI 1.3; $p < .05$). Analyzing breaches by ethnicity, Blacks' and Whites' accounts are most frequently breached. The mean number of recorded exposures associated with the emails for Blacks and Whites is 3.16 and 3.11 respectively. For Hispanics/Latinos, the corresponding number is 2.49 (see Table SI 1.2; $p < .05$). And for Asians, the mean is 2.82.

To assess the source of the exposure, we checked the source of the breaches. The records cover 162 breach identifiers and 156

Table 2: Recorded Exposures by Socioeconomic Factors

	Mean	SE
Age		
18 to 25	1.83	0.09
26 to 35	3.12	0.09
36 to 50	3.34	0.08
51 to 65	3.28	0.07
66 to 100	2.95	0.07
Education		
No HS	2.35	0.12
HS Grad.	2.88	0.06
Some College	3.03	0.09
2-year College Degree	3.07	0.10
4-year College Degree	3.20	0.09
Postgrad Degree	3.20	0.11
Sex		
Female	3.16	0.05
Male	2.81	0.05
Race		
White	3.11	0.05
Black	3.16	0.11
Hispanic/Latino	2.49	0.08
Asian	2.82	0.21
Native American	2.96	0.26
Mixed Race	2.66	0.24
Other	2.45	0.22
Middle Eastern	2.92	1.32

nonempty domains (1,722 records have no domain), but there was a sharp skew with 21 domains with more than 100 records accounting for 11,761 exposures. Table 3 lists the 21 sites. Prominent websites like linkedin.com, adobe.com, dropbox.com, lastfm.com, among others feature on the list.

The analysis so far includes all recorded exposures. All observed records are marked verified (100%); 37.77% are categorized as *Spam-List*. Excluding spam lists leaves 9,304 qualifying records. We retain all 5,000 respondents, including those with zero qualifying records: the mean is 1.86 and 71.78% have at least one qualifying exposure.

When we look at education, the pattern holds up. Once again, the number of recorded exposures per person for people with a college degree or more is higher than for people who only got as far as high school (see Table 4). Moving to sex, the pattern is more attenuated with women just nudging ahead of men—the mean for women and men is 1.87 and 1.85 respectively, with little evidence of a difference. The general pattern for age remains roughly similar to what we saw above, with the middle-aged having higher recorded counts than the youngest and oldest groups. Breaking down by race, we see some interesting changes. The means for Asians and Whites are 2.00 and 1.97. Hispanics/Latinos have a lower mean of 1.51. The mean for Blacks is 1.78, below that for Whites. This change concerns spam-list inclusion; verification status does not vary among the observed records.

Table 3: Most Frequently Implicated Domains

Domain	Records
rivercitymediaonline.com	2911
linkedin.com	1089
modbsolutions.com	1067
myspace.com	1059
data4marketers.com	996
cashcrate.com	854
adobe.com	607
disqus.com	570
ticketfly.com	393
tumblr.com	336
dropbox.com	286
dailymotion.com	255
last.fm	244
evony.com	169
clixsense.com	150
cafemom.com	145
imesh.com	144
kickstarter.com	140
edmodo.com	128
zomato.com	112
neopets.com	106

4 Conclusion

Recorded exposure affects 82.84% of sampled adults. In total, the 5,000 email accounts on file are associated with 14,951 exposure records. The average is 2.99 recorded exposures per person. This number, though, is the lower bound for three reasons. First, not all breaches are made public. Second, HIBP doesn't allow access to data on sensitive breaches—breached online accounts on services that may have reputational consequences for people—via its public API. Third, many Americans have multiple email accounts. We only had one email ID per person.

The demographic patterns are consistent with a role for differences in online-service use, although these data do not establish that mechanism. This finding is consistent with Laohaprapanon and Sood, who find that the better educated, people with higher incomes, and racial majorities spend a smaller proportion of time online on problematic sites, but because they are online more often, they end up visiting more such sites [7]. This is contrary to the traditional narrative about the digital divide [15].

Table 4: Verified, Nonspam Exposures by Socioeconomic Factors

	Mean	SE
Age		
18 to 25	1.41	0.08
26 to 35	2.25	0.08
36 to 50	2.14	0.06
51 to 65	1.88	0.05
66 to 100	1.48	0.05
Education		
No HS	1.32	0.09
HS Grad.	1.64	0.05
Some College	1.98	0.07
2-year College Degree	1.88	0.08
4-year College Degree	2.15	0.07
Postgrad Degree	2.13	0.08
Sex		
Female	1.87	0.04
Male	1.85	0.04
Race		
White	1.97	0.04
Black	1.78	0.08
Hispanic/Latino	1.51	0.06
Asian	2.00	0.18
Native American	1.71	0.18
Mixed Race	1.77	0.19
Other	1.46	0.17
Middle Eastern	2.69	1.19

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